Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 100218 | *Subject: Python Programming* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which of the following is a valid variable name in Python?
(i) 2value (iii) value-2
(ii) value2 (iv) value 2

(b) Which symbol is used for comments in Python?
(i) // (iii) #
(ii) <!-- --> (iv) /* */

(c) Which function is used to take input from the user?
(i) scanf() (iii) read()
(ii) gets() (iv) input()

(d) Which of the following will create an empty dictionary?
(i) {} (iii) dict()
(ii) [] (iv) set()

(e) What is the correct syntax to define a function in Python?
(i) function fun(): (iii) def fun[]:
(ii) def fun(): (iv) define fun():

(f) "Hello"[1] gives:
(i) H (iii) l
(ii) e (iv) o

(g) Which method is used to find length of a string?
(i) length() (iii) count()
(ii) size() (iv) len()

(h) Lists in Python are
(i) Mutable (iii) Constant
(ii) Immutable (iv) None

(i) Which method removes the last element from a list?
(i) remove() (iii) clear()
(ii) pop() (iv) delete()

(j) Which of the following is used to open a file in Python?
(i) file() (iii) read()
(ii) open() (iv) fopen()

Q.2 (a) Explain Identifiers, Keywords, Statements, and Expressions in Python. [7]
(b) Explore the working of while and for loop with examples. [7]

Q.3 (a) Explain Assignment and Relational operators in python with program? [7]
(b) What is type conversion? Explain with example? [7]

Q.4	(a)	Explain elseif ladder in python programming with suitable example.	[7]
	(b)	Write a Python program to count how many numbers in a given list are even and how many are odd.	[7]
Q.5	(a)	Explain the scope and lifetime of variables with examples. Also demonstrate how the global keyword is used to modify global variables inside a function.	[7]
	(b)	Write a function factors (N) that returns a list of all positive divisors of N (N>=1). For example: factors (6) returns [1,2,3,6] factors (1) returns [1] factors (13) returns [1,13]	[7]
Q.6	(a)	Explain concatenation, comparison, and length operation. Support your explanation with suitable example programs.	[7]
	(b)	Explain commonly used string methods such as upper(), lower(), strip(), find(), replace(), split(), join(), etc. Write example code demonstrating at least five string methods.	[7]
Q.7	(a)	Explain operations such as *concatenation*, *repetition*, *membership*, and *length* in list with program?	[7]
	(b)	Explain indexing and slicing in lists with examples. Discuss: Positive and negative indexing, Accessing single elements , Slicing formats: list[start:end], list[start:end:step].	[7]
Q.8	(a)	What are nested loops? Write a program to print the following pattern using nested for loops: * ** *** ****	[7]
	(b)	Explain commonly used methods like add(), update(), remove(), discard(), union() in set with program?	[7]
Q.9	(a)	How to perform read operation in text file explain with example?	[7]
	(b)	Write a Python function to count the frequency of each character in a given a string and return the output in a dictionary. Example: char_frequency ("HELLO") returns {"H":1, "E":1, "L":2, "O":1}	[7]